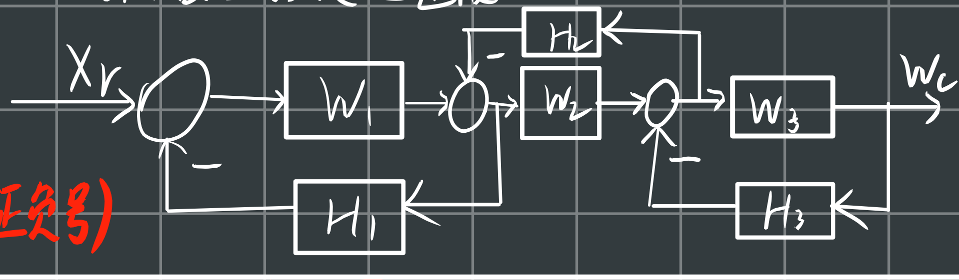
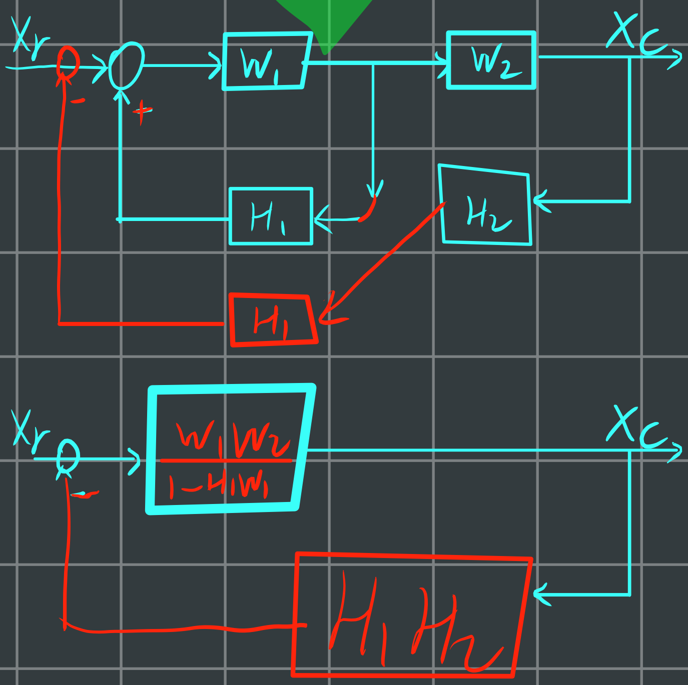
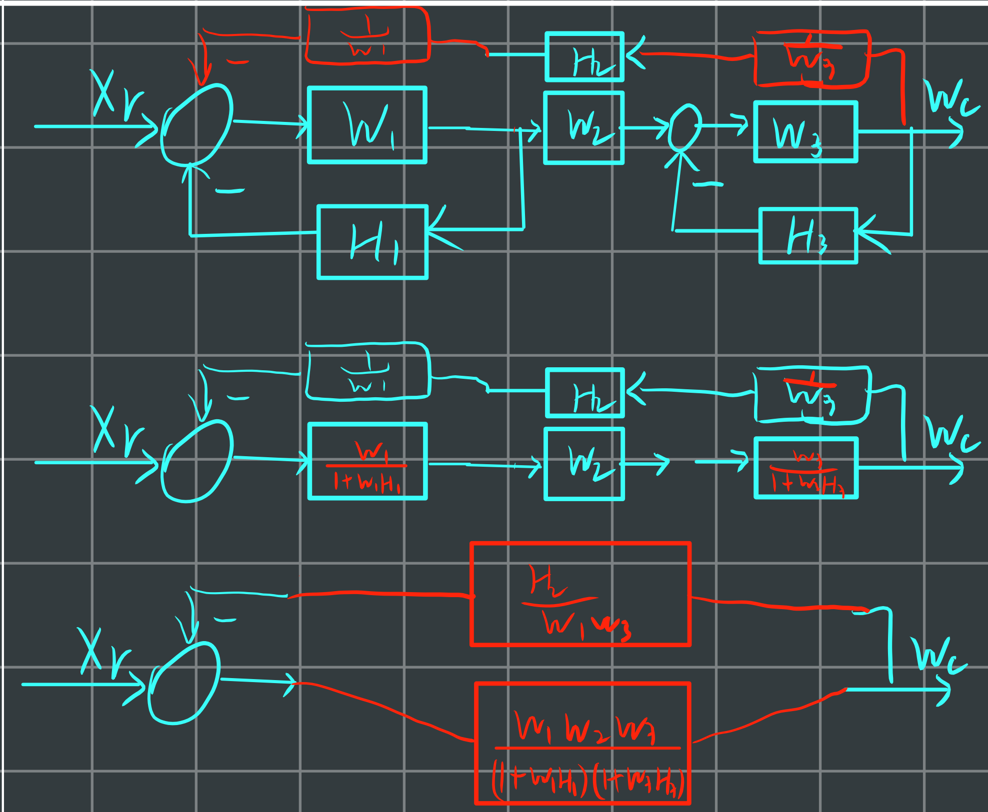
2-25 分别简化两结构图，并求出相应的传递函数

（看清正负号）

$$W_g(s)=\frac{W_1W_2}{1-H_1W_1+W_1W_2H_1H_2}$$

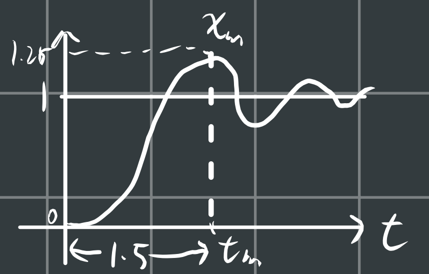
$$W_g(s)=\frac{\dfrac{W_1W_2W_3}{(1+W_1H_1)(1+W_3H_3)}}{1+\dfrac{H_2}{\cancel{W_1W_3}}\cdot\dfrac{\cancel{W_1}W_2\cancel{W_3}}{(1+W_1H_1)(1+W_3H_3)}}$$

$$=\frac{W_1W_2W_3}{1+W_1H_1+W_2H_2+W_3H_3+W_1H_1W_3H_3}$$

3-11 一个单位反馈控制系统的开环传递函数为

$W_k(S)=\frac{\omega_n}{S(S+2\xi\omega_n)}$，其单位阶跃响应曲线如图，图中 $X_m=1.25$

$t_m=1.5S$，试确定系统参数 K_k 及 τ 值



题 3-11 一单位反馈控制系统的开环传递函数为

$$W_K(s)=\frac{K_K}{s(\tau s+1)}$$

其单位阶跃响应曲线如图 P3-1 所示，图中的 $X_m=1.25$，$t_m=1.5\text{s}$。试确定系统参数 K_K 及 τ 值。

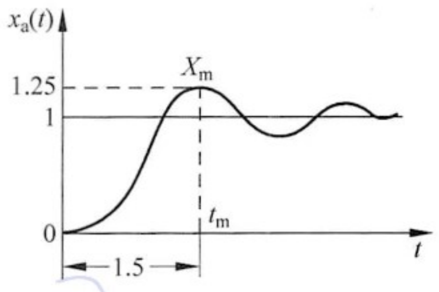


图 P3-1 题 3-11 系统的单位阶跃响应曲线

解 由图 P3-1 可知

$$\begin{cases} t_m=\dfrac{\pi}{\omega_n\sqrt{1-\xi^2}}=1.5 \\ \sigma\%=\dfrac{x_m-x(\infty)}{x(\infty)}\times100\%=\dfrac{1.25-1}{1}\times100\%=25\%=e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}}\times100\% \end{cases}$$

解得

$$\begin{cases} \xi\approx0.4 \\ \omega_n\approx2.285 \end{cases}$$

本系统的开环传递函数整理为

$$W_K(s)=\frac{\frac{K_K}{\tau}}{s\left(s+\frac{1}{\tau}\right)}$$

与标准形式 $W_k(s)=\frac{\omega_n^2}{s(s+2\xi\omega_n)}$ 相对比，得

$$\begin{cases} \dfrac{K_K}{\tau}=\omega_n^2=2.285^2 \\ \dfrac{1}{\tau}=2\xi\omega_n=2\times0.4\times2.285 \end{cases}$$

解得

$$\begin{cases} K_K\approx2.856 \\ \tau\approx0.547 \end{cases}$$

3-13已知单位反馈控制系统的开环传递函数为 $W_k(s)=\dfrac{K_k}{S(\tau S+1)}$ 试选择Kk及τ值以满足下列指标:

(1)当xr(1)=t时，系统的稳态误差ev(∞)≤0.02;

(2)当xr(t)=1(t)时，系统的σ%≤30% ,ts(5%)≤0.3s.

e(∞) \ xr	1(t)	t	$\frac{1}{2}t^2$
0	$\frac{1}{1+K_k}$	∞	∞
I	0	$\frac{1}{K_k}$	∞
II	0	0	$\frac{1}{K_k}$

$\frac{1}{K_k} \le 0.02$ $\quad K_k \ge 50$

解 $W_B(s) = \frac{W_k(s)}{1+W_k(s)} = \frac{\frac{K_k}{\tau}}{s^2+\frac{1}{\tau}s+\frac{K_k}{\tau}}$ $\quad W_B(s) = \frac{\omega_n^2}{s(s+2\xi\omega_n)}$

$\sigma\% = e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}} \le 0.3$ $\quad \xi \ge 0.358$

$t_s(5\%) = \frac{3}{\xi\omega_n} \le 0.3$ $\quad \omega_n \ge 27.9$

$$\begin{cases} \frac{K_k}{\tau} = \omega_n^2 \\ \frac{1}{\tau} = 2\xi\omega_n \end{cases}$$

3-19有闭环系统的特征方程式如下，试用劳斯判据判断系统的稳定性,并说明特征根在复平面上的分布。

(5) $s^6+3s^5+9s^4+18s^3+22s^2+12s+12=0$

s^6	1	9	22	12
s^5	3	18	12	
s^4	$\frac{3\times9-18\times1}{3}=3$	$\frac{3\times22-12\times1}{3}=18$	12	
s^3	12	36		
s^2	9	12		
s^1	20			
s^0	12			

$f(s)=3s^4+18s^2+12$

$f'(s)=12s^3+36s$

答：因为第一列无变号，所以S右半平面没有特征根。

又∵ S^3行各系数为零

∴有4个纯虚根，系统临界稳定

3-26 有一个单位反馈系统，其开环传递函数为Wk(s)= $\frac{3s+10}{s(5s-1)}$ 求系统的动态误差系数;

10分

并求当输人量xr(t)=1+t+ $\frac{1}{2}t^2$ 时，稳态误差的时间函数e(t)。

例3-10

P126

系统给定误差传递函数为

$$\frac{E(s)}{X_r(s)} = \frac{5s^2-s}{5s^2+2s+10} = \frac{-s+5s^2}{10+2s+5s^2}$$

$$= -\frac{1}{10}s + \frac{13}{25}s^2 - \frac{27}{500}s^3 + \cdots$$

$$10+2s+5s^2 \overline{\big) -s+5s^2} \quad \text{商: } -\frac{1}{10}s + \frac{13}{25}s^2 - \frac{27}{500}s^3$$

$$-s - \frac{1}{5}s^2 - \frac{1}{2}s^3$$

$$+\frac{26}{5}s^2 + \frac{1}{2}s^3$$

$$\frac{26}{5}s^2 + \frac{26}{25}s^3 + \frac{13}{5}s^4$$

$$-\frac{27}{50}s^3 - \frac{13}{5}s^4$$

$$\therefore k_0 = \infty;\ k_1 = -10;\ k_2 = \frac{25}{13}$$

$$e(t) = \frac{1}{k_0}X_r(t) + \frac{1}{k_1}X_r'(t) + \frac{1}{k_2}X_r''(t) + \cdots\cdots$$

$$X_r(t) = 1 + t + \frac{1}{2}t^2$$

$$X_r'(t) = 1 + t$$

$$X_r''(t) = 1$$

$$X_r'''(t) = 0$$

$$\Rightarrow e(t) = 0 + \left(-\frac{1}{10}\right)(1+t) + \frac{13}{25}$$

$$= -\frac{t}{10} + \frac{21}{50}$$

4-8求下列各开环传递函数所对应的负反馈系统根轨迹。

(1) $W_k(S)=\frac{K_g(S+2)}{S^2+2S+3}$

(2) $W_k(S)=\frac{K_g}{S(S+2)(S^2+2S+2)}$

(3) $W_k(S)=\frac{K_g(S+2)}{S(S+3)(S^2+2S+2)}$ P162 图4-18

(1)

① 分母的解为起点

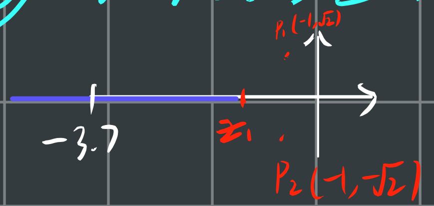
$P_1=-1+j\sqrt{2}$; $P_2=-1-j\sqrt{2}$

② 分子解为极点

$z_1=-2$; 无穷极点

极点与起点个数

③ 在实轴上的根轨迹 (右侧 和为奇数个)



④ 分离点、会合点

$D'(S)N(S)-D(S)N'(S)=0$

$(S^2+2S+3)-(S+2)(2S+2)=0$

$s^2+4s+4=3$

$S_1=-2+\sqrt{3}$; $S_2=-2-\sqrt{3}$ ≈ -0.3(舍)

≈ -3.7

不在根轨迹上实轴

⑤ 渐近线

$\varphi=\frac{\pm180^\circ(1+2M)}{m-n}$ $2-1=1$

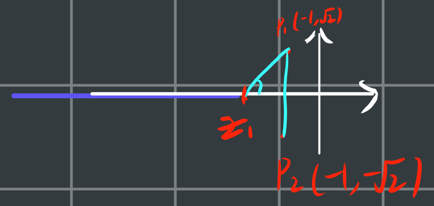
∵ $m-n=1$ ⇒ 不用管？

渐近线交点: $\sigma_k=\frac{\sum_{i=1}^{n}(P_i)-\sum_{i=1}^{m}(z_i)}{n-m}$

⑥ 出射角、入射角

$(\alpha_1+\alpha_2+\dots)-(\beta_1+\beta_2+\dots)=\pm180^\circ(1+2M)$

极点与该点的夹角　　起点与该点夹角

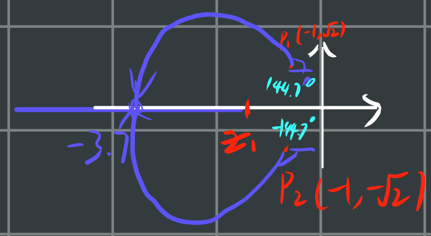


$54.7^\circ-(90^\circ+\beta_{sc})=\pm180(1+2M)$

$k_1-k_2=1$

$\beta_{sc}=\pm180(1+2M)-35.3$

$=144.7^\circ, -35.3^\circ$



(2) $W_k(s)=\dfrac{K_g}{s(s+2)(s^2+2s+2)}$

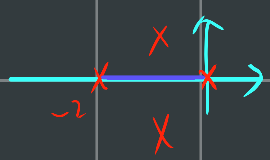
① 起点

$z_1=0,\ z_2=-2,\ z_{3,4}=-1\pm j$

② 零点

无穷零点

③ 实轴上根轨迹 $(-2,0)$



④ 分离点、汇合点

$(2s+2)(s^2+2s+2)+(s^2+2s)(2s+2)=0$

$s^3+3s^2+3s+1=0 \Rightarrow s_{1,2,3}=-1$

⑤ 出射角

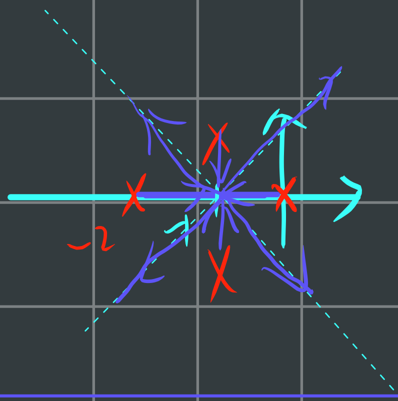
$0-(45^\circ+90^\circ+135^\circ+\beta_{sc})=\pm180(1+2M)$

$\beta_{sc}=90^\circ$

⑥ 渐近线

$$\varphi=\frac{\pm180(1+2u)}{m-n}=\pm45^\circ,\pm135^\circ$$

$$a_k=\frac{-2-(-1-1)}{4}=-1$$



(3) $W_k(s)=\dfrac{k_s(s+2)}{s(s+3)(s^2+2s+2)}$

$\dfrac{kg(s+1)}{s(s+3)(s^2+2s+2)}$

① 起点

$P_1=0,P_2=-3,P_{3,4}=-1\pm j$

② 终点

$Z_1=-2$;无穷

③ 实轴根轨迹

$(-\infty,-3)\cup(-2,0)$

④ 出射角

$45^\circ-(26.6^\circ+135+90^\circ+\beta_{sc_1})=\pm180(1+2u)$

$\beta_{sc_1}=\pm180(1+2u)-26.6^\circ$

$=153.4^\circ、-26.6^\circ$

(选取受渐近线影响?)

? ? ? ? ? ?

⑤ 渐近线

$$\varphi=\frac{\pm180(2u+1)}{m-n}=\frac{\pm180^\circ(1+2u)}{4-1}=\pm60^\circ$$

$$\sigma_r=\frac{\sum P_i-\sum Z_i}{m-n}=-1$$

⑥ 分离点、汇合点

$$\frac{k_g(s+2)}{s(s+3)(s^2+2s+2)}$$

$$s(s+3)(s^2+2s+2)' - (s+2)(s+3)'(s^2+2s+2)$$

$$-(s^2+3s)(+s^2+4s+2)-(2s^2+7s+6)(s^2+2s+2)$$

$$(3s^2+10s+6)(s^2+2s+2)+2s^3+6s^2=0$$

$$3s^4+18s^3+38s^2+32s+12=0$$

全虚根

⑦ 与虚轴交点

$$s(s+3)(s^2+2s+2)+k_g(s+2)=0$$

$$s^4+5s^3+8s^2+(6+k_g)s+2k_g=0$$

由劳氏判据得

s^4	1	8	$+2k_g$
s^3	5	$(6+k_g)$	
s^2	$\frac{34-k_g}{5}$	$2k_g$	
s^1	$(6+k_g)-\left(\frac{50k_g}{34-k_g}\right)$		
s^0	$2k_g$		

令 $s^1=0 \Rightarrow k_g=7$

$$\Rightarrow F(s)=\frac{27}{5}s^2+14=0$$

$$\Rightarrow s=j\sqrt{\frac{70}{27}}$$

$$\approx 1.61j$$

5-8绘出下列各传递函数对应的幅相频率特性

(5) $W(s)=\frac{4}{s(s+2)}$

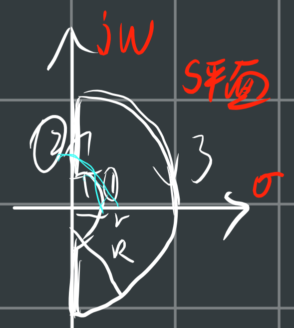
(6) $W(s)=\frac{4}{(s+1)(s+2)}$

$\frac{10}{s(s-3)}$

$\frac{10}{(s+1)(s+2)}$

(5)

$$W_k(s)=\frac{2}{s\left(\frac{1}{2}s+1\right)}$$



$R=\infty$

$r=0^+$

① $s=\rho e^{j\theta},\ \rho=0^+,\ \theta=0^\circ\to 90^\circ$

$$W_k(\rho e^{j\theta})=\frac{2}{\rho e^{j\theta}\left(\frac{1}{2}\rho e^{j\theta}+1\right)}=\frac{2}{0^+e^{j\theta}\left(\cancel{\frac{1}{2}0^+e^{j\theta}}+1\right)}$$

太小，忽略不记

$$|W_k(\rho e^{j\theta})|=+\infty$$

$$\angle W_k(\rho e^{j\theta})=0^\circ\xrightarrow{逆}-90^\circ$$

② $s=j\omega,\quad \omega=0^+\to+\infty$

$$W_k(j\omega)=\frac{2}{j\omega\left(\frac{1}{2}j\omega+1\right)}$$

$$|W_k(j\omega)|=\frac{2}{\omega\sqrt{\frac{1}{4}\omega^2+1}}=+\infty\to 0^+$$

$$\angle W_k(j\omega)=0-\arctan\frac{\omega}{0}-\arctan\frac{\frac{1}{2}\omega}{1}$$

$$=-90^\circ\to-180^\circ$$

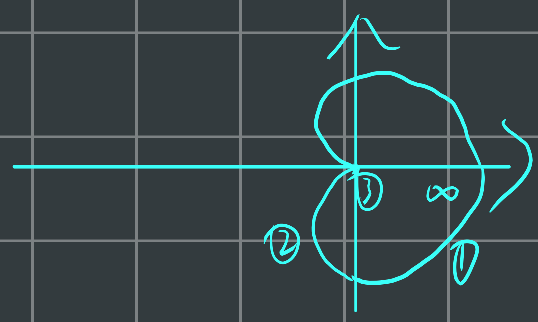
③ $s=\rho e^{j\theta},\ \rho=+\infty,\ \theta=90^\circ\to 0^\circ$

$$W_k(\rho e^{j\theta}) = \frac{2}{\rho e^{j\theta}(\frac{1}{2}\rho e^{j\theta}+1)} = \frac{2}{\infty e^{j\theta}(\frac{1}{2}\infty e^{j\theta}+1)} \approx \frac{2}{\infty e^{2j\theta}} = 0^+ e^{-2j\theta}$$

$$|W_k(\rho e^{j\theta})| = 0^+$$

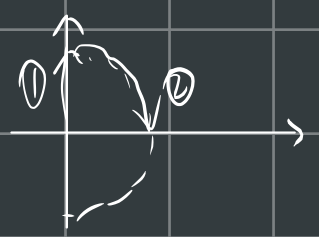
$$\angle W_k(\rho e^{j\theta}) = -180^\circ - 0$$

$$Z = P - N = 0$$



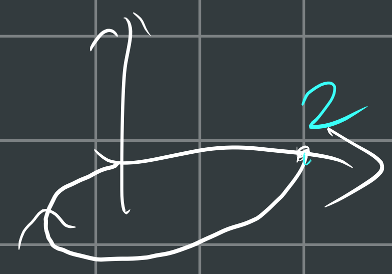
∴ 2才稳

(2)
$$W_k(s) = \frac{4}{(s+1)(s+2)} = \frac{2}{(s+1)(\frac{s}{2}+1)}$$



① $s = jw$, $w = 0^+ \to +\infty$

$$W_k(jw) = \frac{2}{(jw+1)(\frac{jw}{2}+1)}$$



$$|W_k(jw)| = \frac{2}{\sqrt{w^2+1}\sqrt{(\frac{w}{2})^2+1}} = 2 \to 0$$

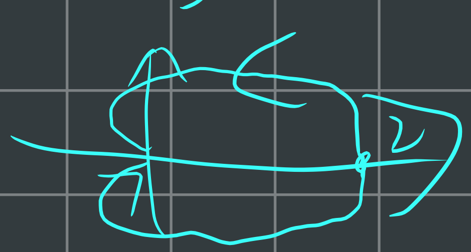
$$\angle W_k(jw) = 0^\circ - \arctan\frac{w}{1} - \arctan\frac{w}{2} = 0^\circ \to -180^\circ$$

② $s = \rho e^{j\theta}$, $\rho = +\infty$, $\theta = 0^\circ \to 90^\circ$

$$W_k(\rho e^{j\theta}) = \frac{2}{(\rho e^{j\theta}+1)(\frac{\rho e^{j\theta}}{2}+1)} = \frac{2}{+\infty e^{2j\theta}} = 0^+ e^{-2j\theta}$$

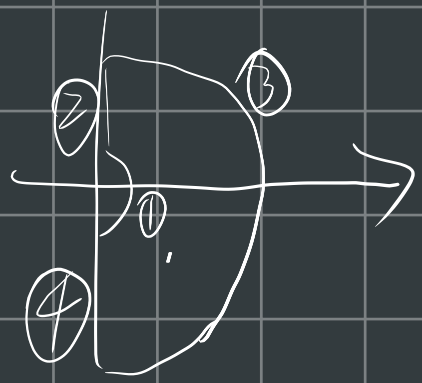
$|\sim| = 0^+$, $\angle W_k(\rho e^{j\theta}) = 0^\circ \to -180^\circ$

所以



5—11用奈氏稳定判据判断下列反馈系统的稳定性，各系统开环传递函数如下：

(2) $W_K(S)=\dfrac{10}{S(S-1)(0.2S+1)}=\dfrac{-10}{S(S+1)(0.2S+1)}$



① $S=\rho e^{j\theta}$, $\rho=0^+$

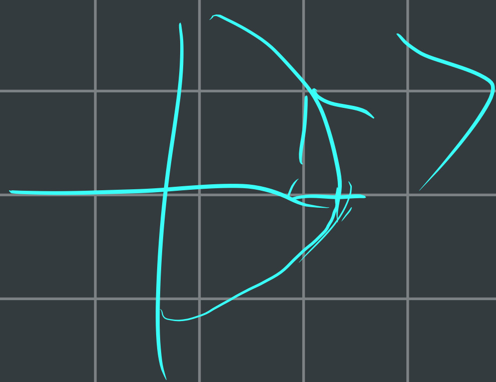
$\theta=-90^\circ\to 90^\circ$

$$W_K(\rho e^{j\theta})=\frac{-10}{\rho e^{j\theta}(\rho e^{j\theta}+1)(0.2\rho e^{j\theta}+1)}$$

$$=\frac{-10}{0^+ e^{j\theta}(\cancel{0^+ e^{j\theta}}+1)(\cancel{0^+ e^{j\theta}}+1)}$$

$$=-\infty e^{-j\theta}$$

$|W_K(\sim)|=\infty$



$\angle W_K(\rho e^{j\theta})=-90^\circ\to 90^\circ$

② $S=j\omega$, $\omega=0^+\to+\infty$

$$W_K(j\omega)=\frac{-10}{j\omega(1-j\omega)(0.2j\omega+1)^2}$$

$$|W_K(j\omega)|=\frac{|-10|}{\sqrt{(j\omega)^2}\ \sqrt{1^2+(j\omega)^2}\ \sqrt{(0.2j\omega)^2+1}}=$$

$+\infty\to 0$